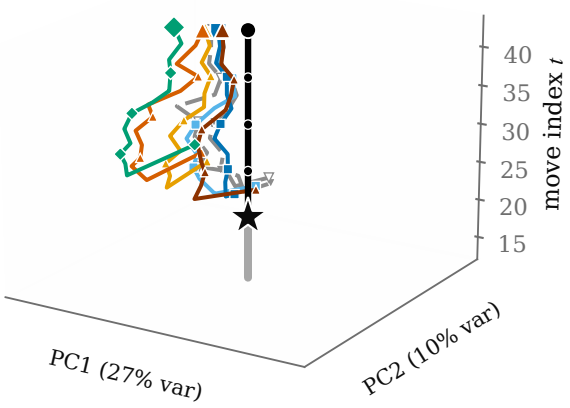
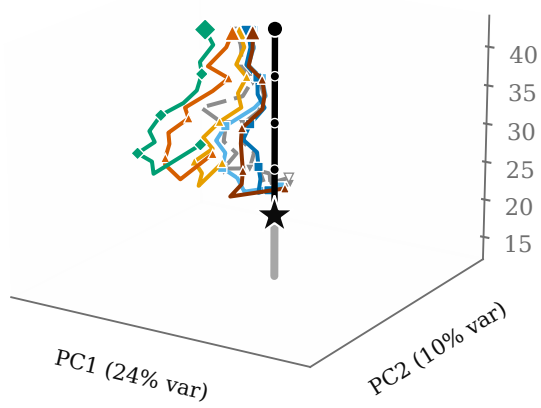
(a) where the nine branches go: the fork in each representation space

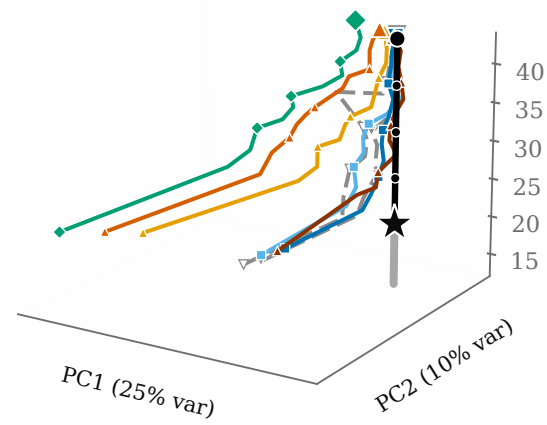
raw layer 5 (reference space)



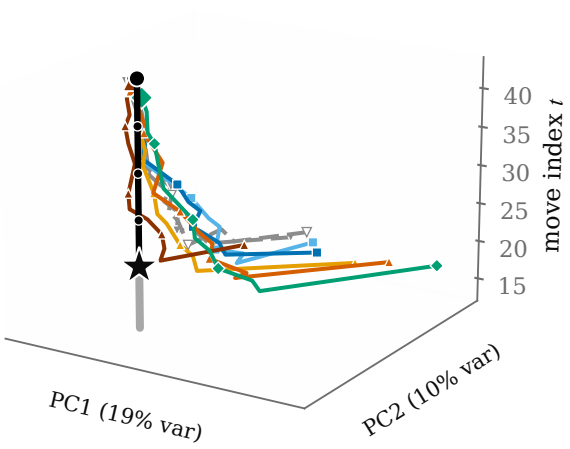
PAnneal (p-anneal, pen 0.042)



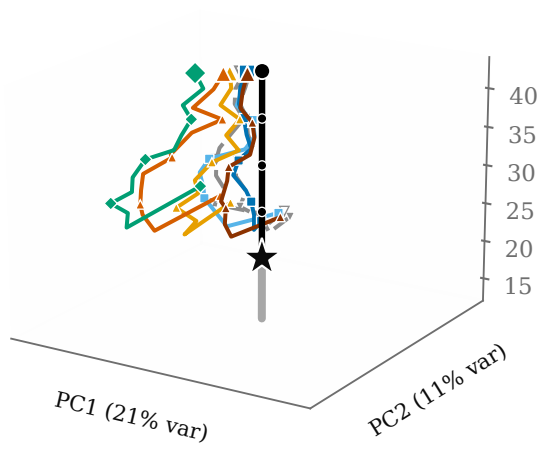
Standard (L1 0.059)



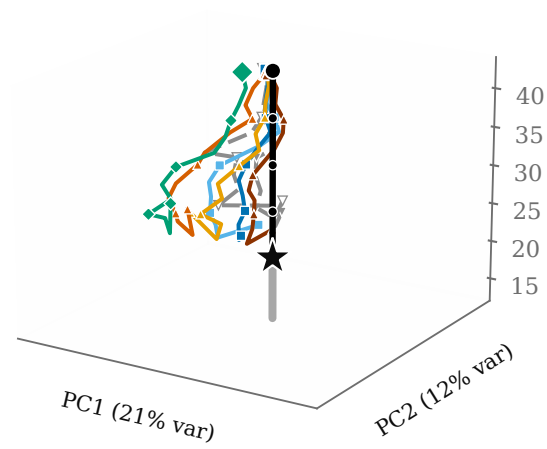
GatedSAE (L1 1.37)



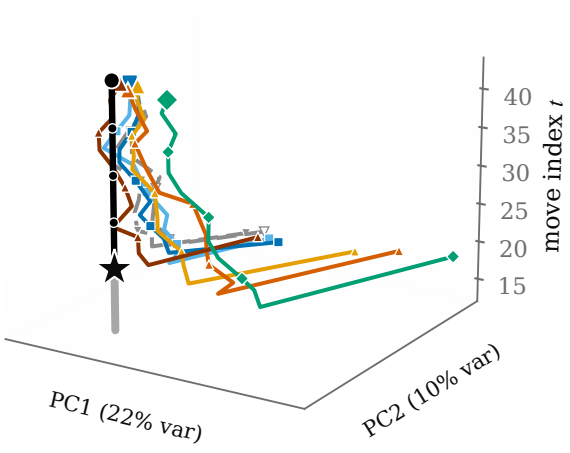
TopK (k=80)



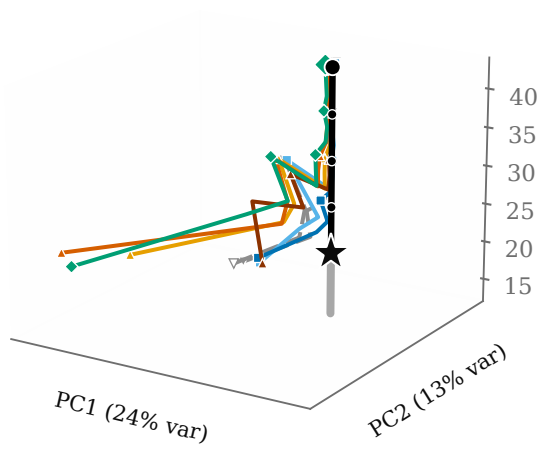
GatedAnneal (p-anneal, pen 0.955)



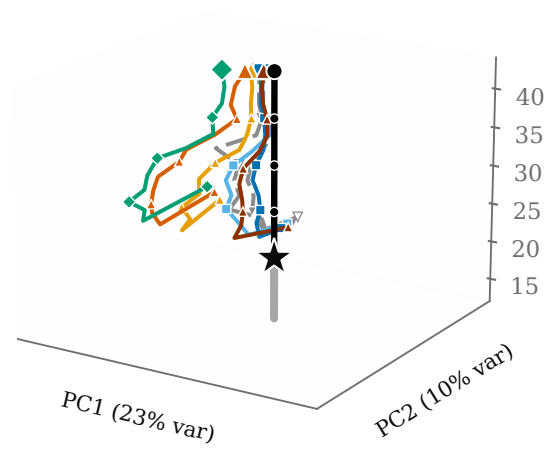
BatchTopK (k=80)



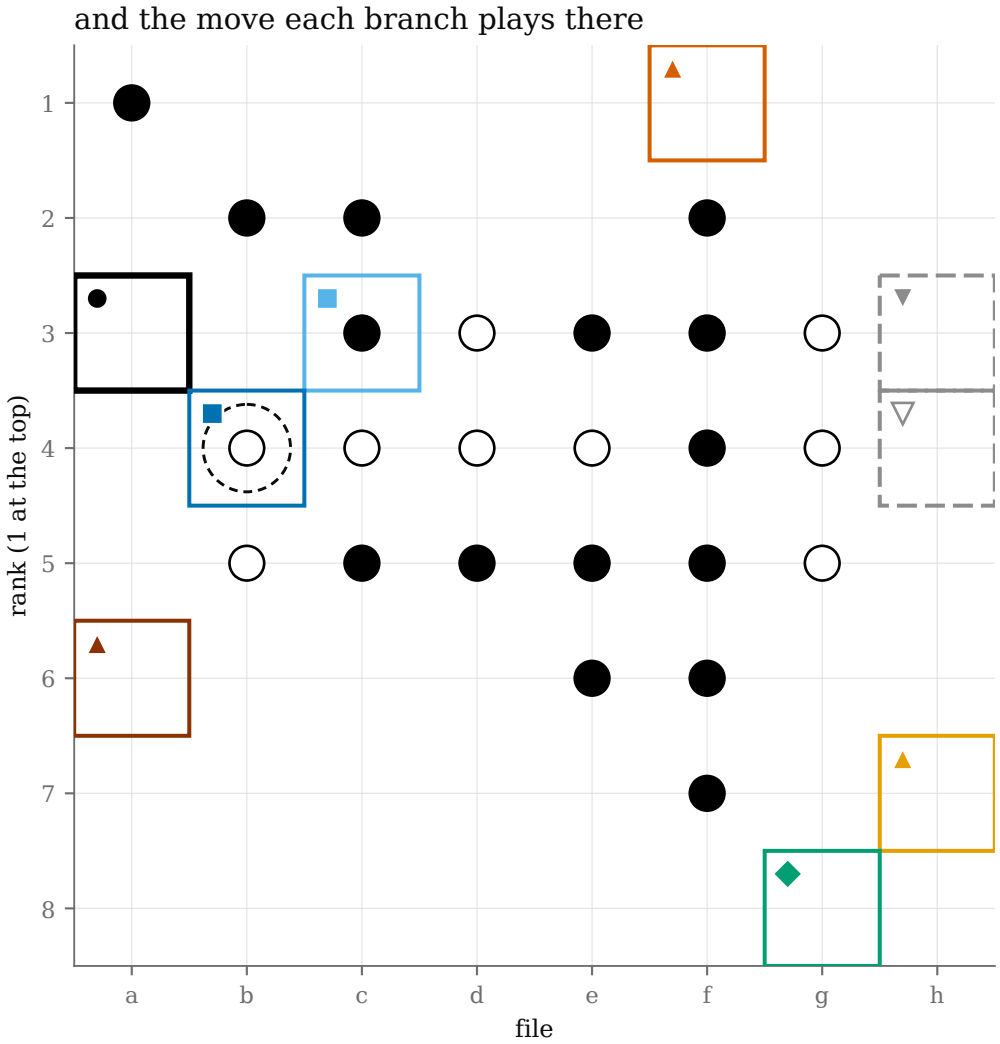
CosineSpaDE (γ=1.0)



SpaDE (λ=1e-3, γ=1e-4)



(b) what they are: the position at t_0,
and the move each branch plays there



the same nine branches in both panels, game 1235, t_0=20
(a) PCA plane refit per space — arb. units, not comparable across cells;
view elev=18, azim=-58; every 2nd ply
(b) 10 moves are legal at t_0; the played move and two of the other 9 are branched

one game (n=1): an illustration of the design, not a claim — the study's numbers, with their CIs over 13,531 branches, are in the parent README
game 1235 is SELECTED, not a typical draw: it is the argmax, over all 1550 branch-complete games, of the mutual separation of its six illegal displacement trajectories, scored in raw layer 5 in the full 512-dimensional space and never in an SAE space or a 2-D projection. It is the clearest available illustration of the construction, not a typical one (population median 0.705, this game 1.080).

- shared prefix (branches identical)
- reference (the game as played)
- R1 occupied (own)
- R1 occupied (opp)
- R2 detached
- R2 own-only
- R2 no bracket
- R3 opponent-only
- branch point t_0
- control: another legal move (null)
- control 2: a third legal move (null)

discs in absolute colour, black to move at t_0; outlined cell = a branch square, its marker in the corner so a disc underneath stays visible; dashed ring = a disc the played move flips

R1 placement — a move puts one disc of the mover's color on an empty square.
R2 bracketing — the disc must close a gapless line of opponent discs with one of the mover's own.
R3 turn — colors alternate; legality is judged for the mover, not the opponent.